EE 3EY4
Lab 3

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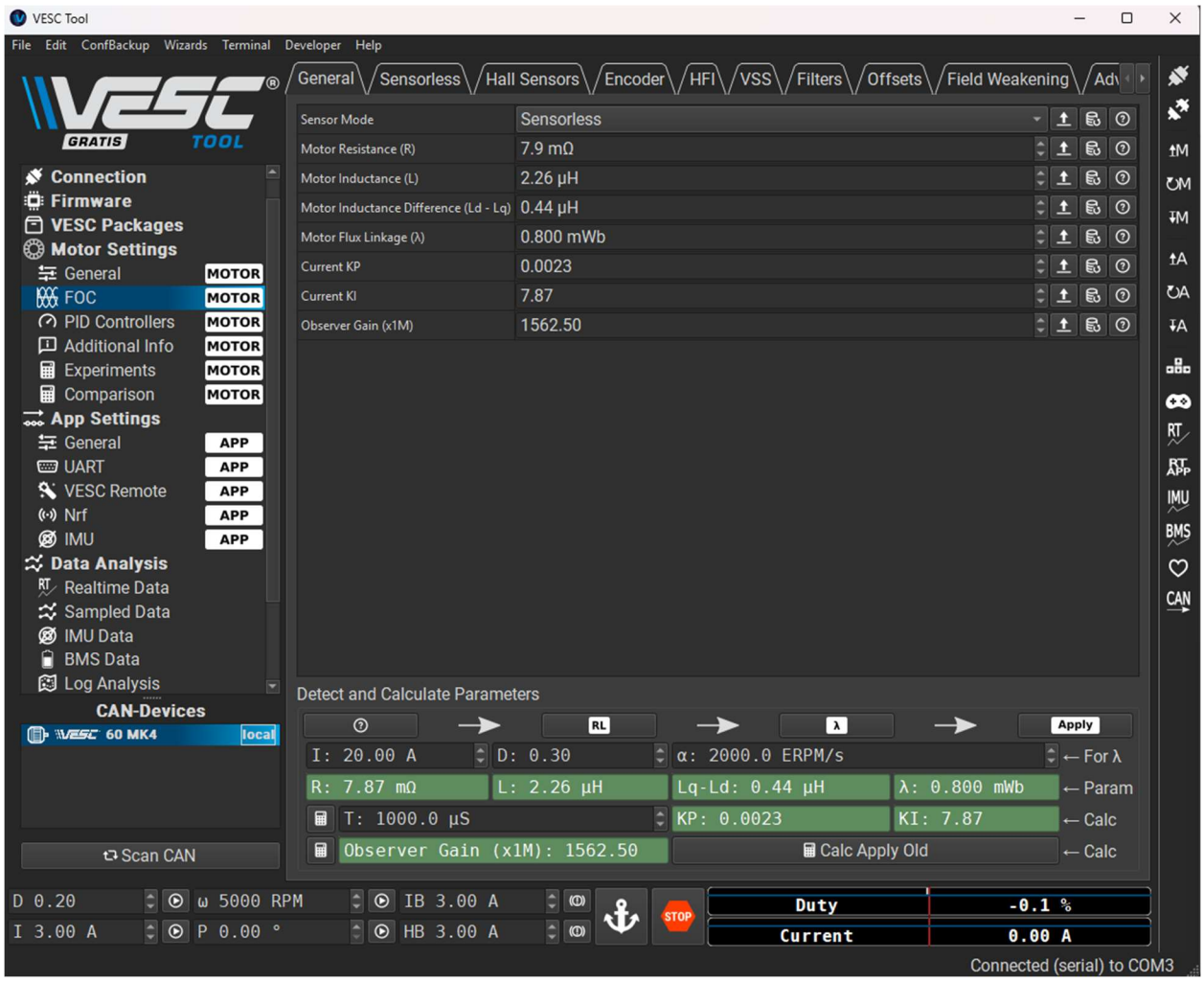
Objective 3: Led by Ryan & Ian

Question 1: What do those parameters represent? Why are those parameters essential to control the motor?



Stator Resistance (R) and Self-Inductance (L)

The parameters **R** and **L** represent the stator resistance and self-inductance per phase. These are fundamental for accurately modeling the relationship between voltage and current within the motor. Resistance **R** is used to calculate the Ohmic voltage drop within the q-axis voltage equation, which is critical for the direct control of torque. Self-inductance **L** is utilized to compensate for the phase lag of the torque vector caused by motor velocity and current. This allows the system to apply the correct d-axis voltage, enabling the motor to reach higher speeds through effective flux control.

Inductance Difference ($L_q - L_d$) and Flux Linkage (λ)

The parameters $\mathbf{L_q - L_d}$ and $\boldsymbol{\lambda}$ represent the saliency (the difference between q-axis and d-axis inductances) and the permanent magnet flux linkage, respectively. Both are required to model

how d/q-axis currents translate into output torque. The flux linkage $\boldsymbol{\lambda}$ relates the back EMF to the motor speed and defines the torque produced by the q-axis current. In Interior Permanent Magnet Synchronous Motors (IPMSMs), the inductance difference $\mathbf{L_q - L_d}$ generates reluctance torque, which improves overall energy efficiency and enables sensorless control through saliency detection.

PID Control Parameters (K_p and K_i)

In Field-Oriented Control (FOC), $\mathbf{K_p}$ (proportional gain) and $\mathbf{K_i}$ (integral gain) are the primary PID parameters used to regulate d/q-axis voltages based on the error between desired and measured currents.

- $\mathbf{K_p}$: Acts proportionally to the current error to determine the speed of the system's corrective response.
- $\mathbf{K_i}$: Operates on the integral of the error over time, serving to eliminate steady-state error and correct for cumulative offsets.

Observer Gain

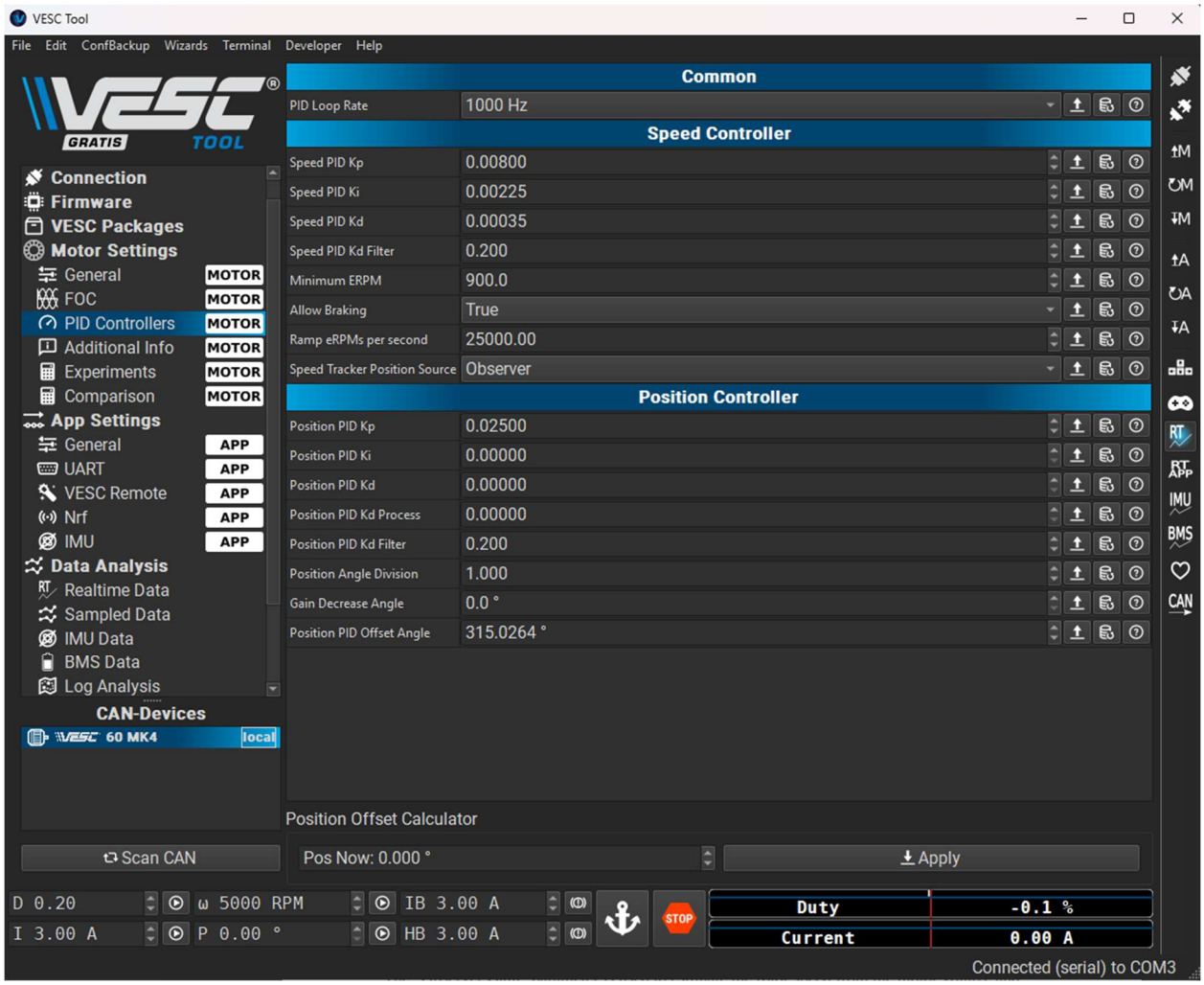
The **Observer Gain** is used by the VESC to estimate rotor speed solely from voltage and current measurements, facilitating sensorless motor control. While increasing the Observer Gain improves the accuracy and tracking of the speed estimate, it simultaneously reduces the system's stability and increases its sensitivity to electrical noise and external disturbances.

<u>Objective 4: Led by Ryan & Ian</u>

<u>Question 2</u>: Explain the importance of PID speed control parameters in relation to the operation of McMaster AEV?

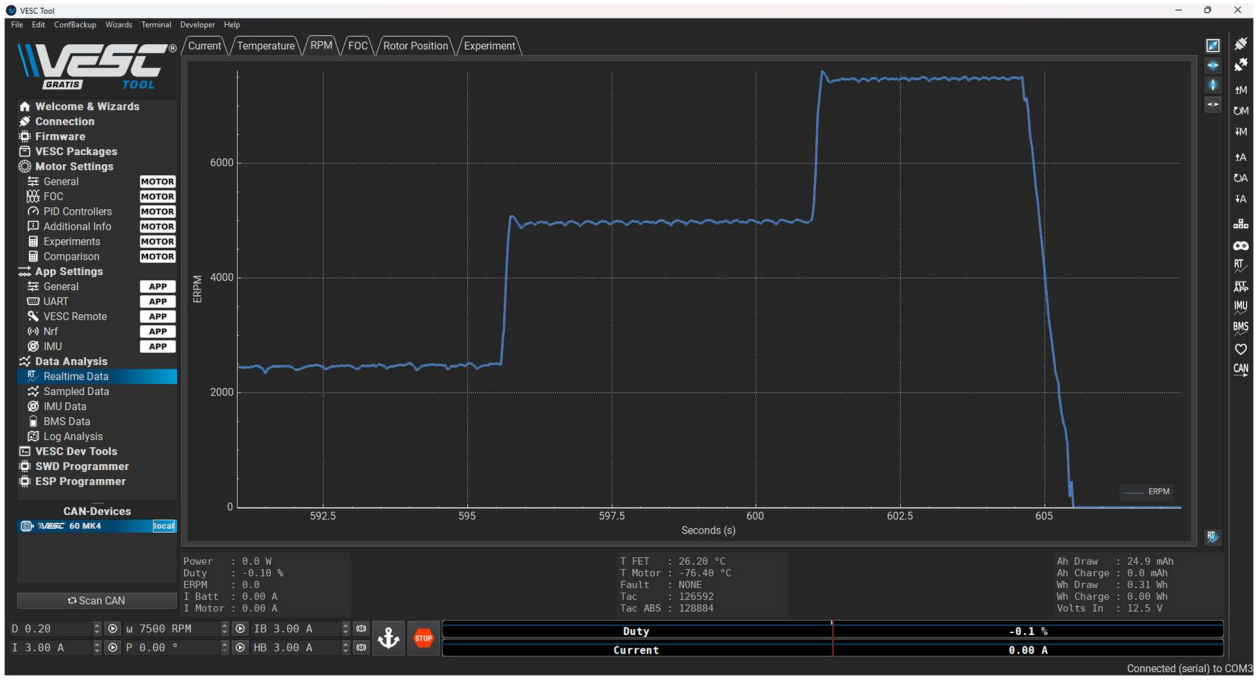
The PID speed control parameters are important because they directly impact how the vehicles move under different conditions. This determines the accuracy and safety of operating the vehicle. K_p needs to be correctly tuned so that the speed of the vehicle stays stable, affecting the acceleration and deceleration, which could lead to friction, oscillations, or jerky motion. On the other hand, K_i eliminates steady state errors that occur in the real world such as rolling resistance and maintaining speed over long periods of time.

By operating both K_p and K_i together, we can ensure correct speeds over long periods of time, reduce mechanical stress, and reproduce test results.



<u>Question 3:</u> First apply speed command to VESC and record how the speed, current and voltage change in time. Please change the speed gradually to a maximum speed command of 7500 rpm. (Recommended: start with a speed of 2500, then 5000 and finally 7500 rpm, Take a screenshot of Your RPM and FOC graphs. Log your data for each speed point and plot the graphs for RPM, FOC current (d- and q-axis currents), FOC voltage (d- and q-axis voltages).

RPM Change:


VESC Tool
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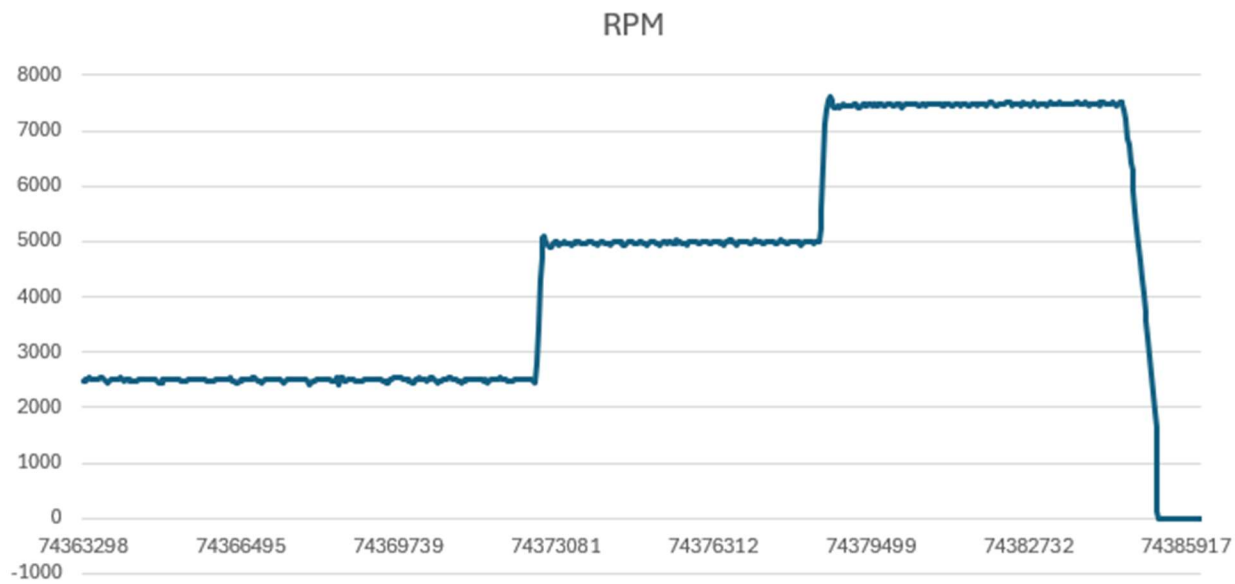
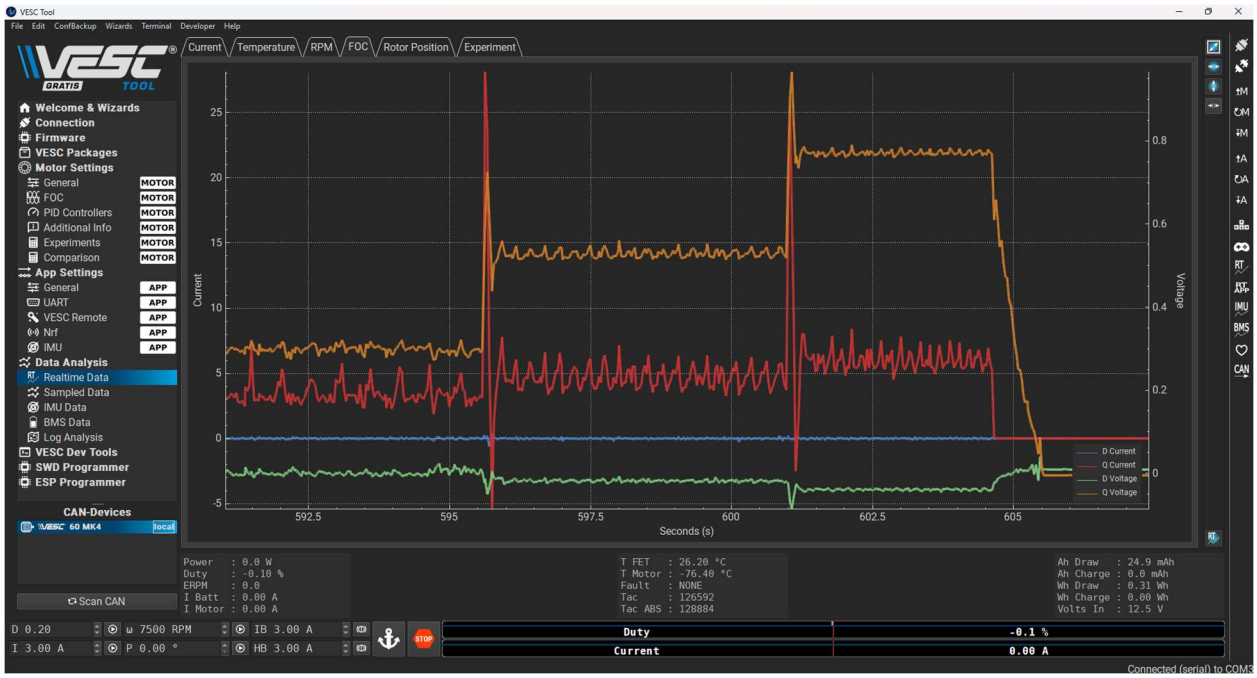
IMU Data
BMS Data
Log Analysis
VESC Dev Tools
SWD Programmer
ESP Programmer
CAN-Devices
Scan CAN
ERPM
Seconds (s)
Power : 0.0 W
Duty : -0.10 %
ERPM : 0.0
I Batt : 0.00 A
I Motor : 0.00 A
T FET : 26.20 °C
T Motor : -76.40 °C
Fault : NONE
Tac : 126592
Tac ABS : 128884
Ah Draw : 24.9 mAh
Ah Charge : 0.0 mAh
Wh Draw : 0.31 Wh
Wh Charge : 0.00 Wh
Volts In : 12.5 V
Duty
-0.1 %
Current
0.00 A
Connected (serial) to COM3


RPM
8000
7000
6000
5000
4000
3000
2000
1000
0
-1000
74363298
74366495
74369739
74373081
74376312
74379499
74382732
74385917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Seconds (s)
D Current
Q Current
D Voltage
Q Voltage
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Duty : -0.10 %
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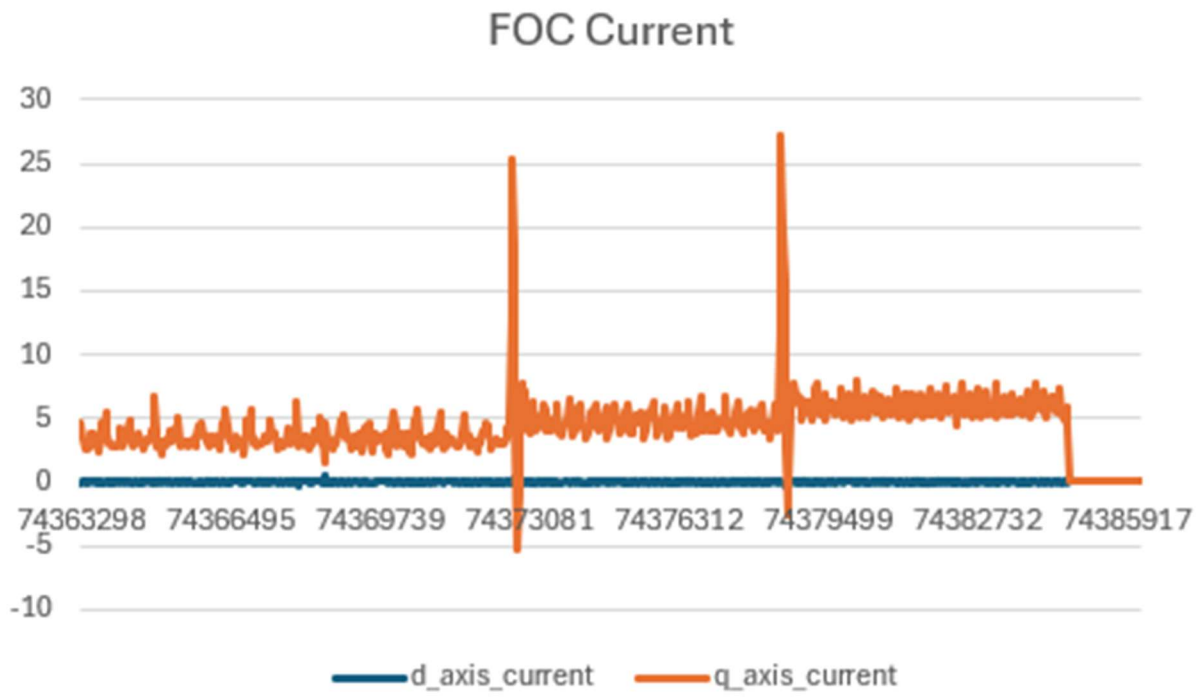
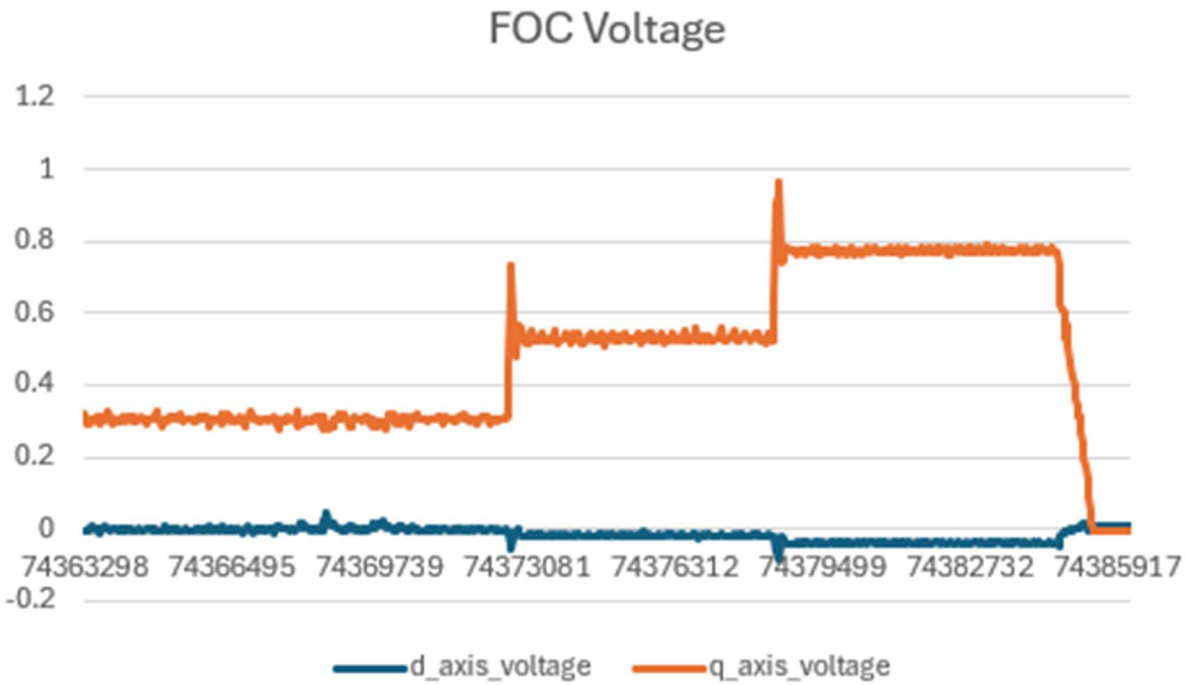
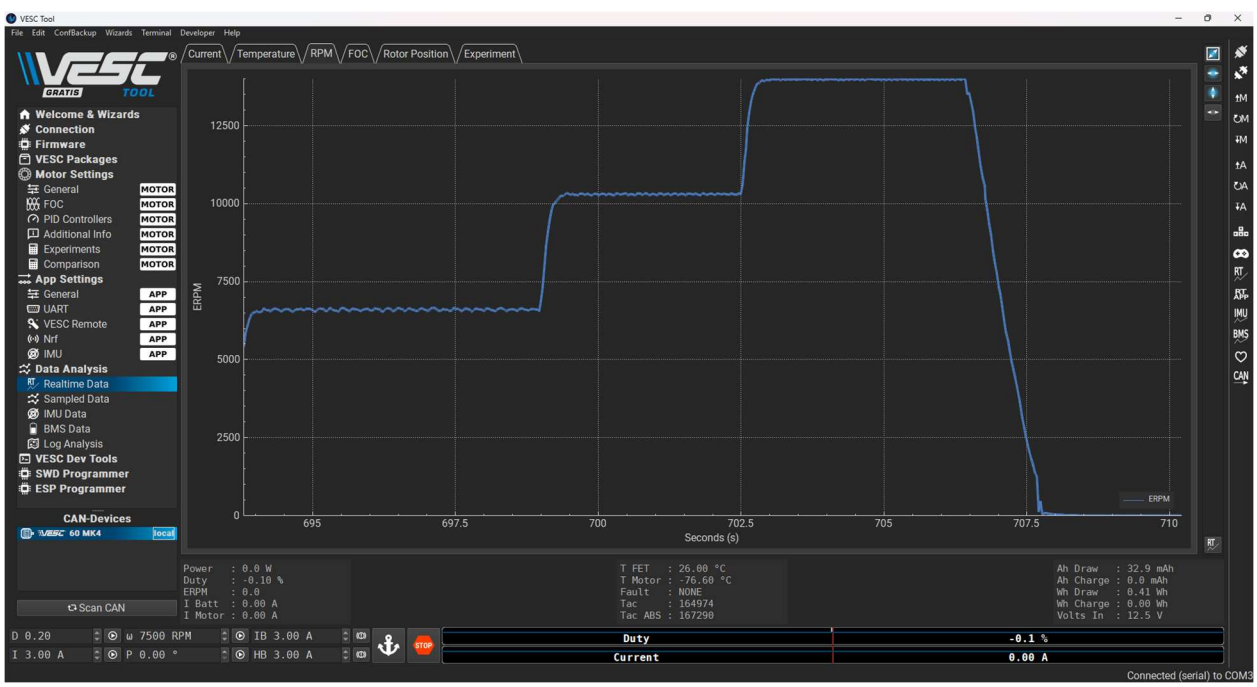
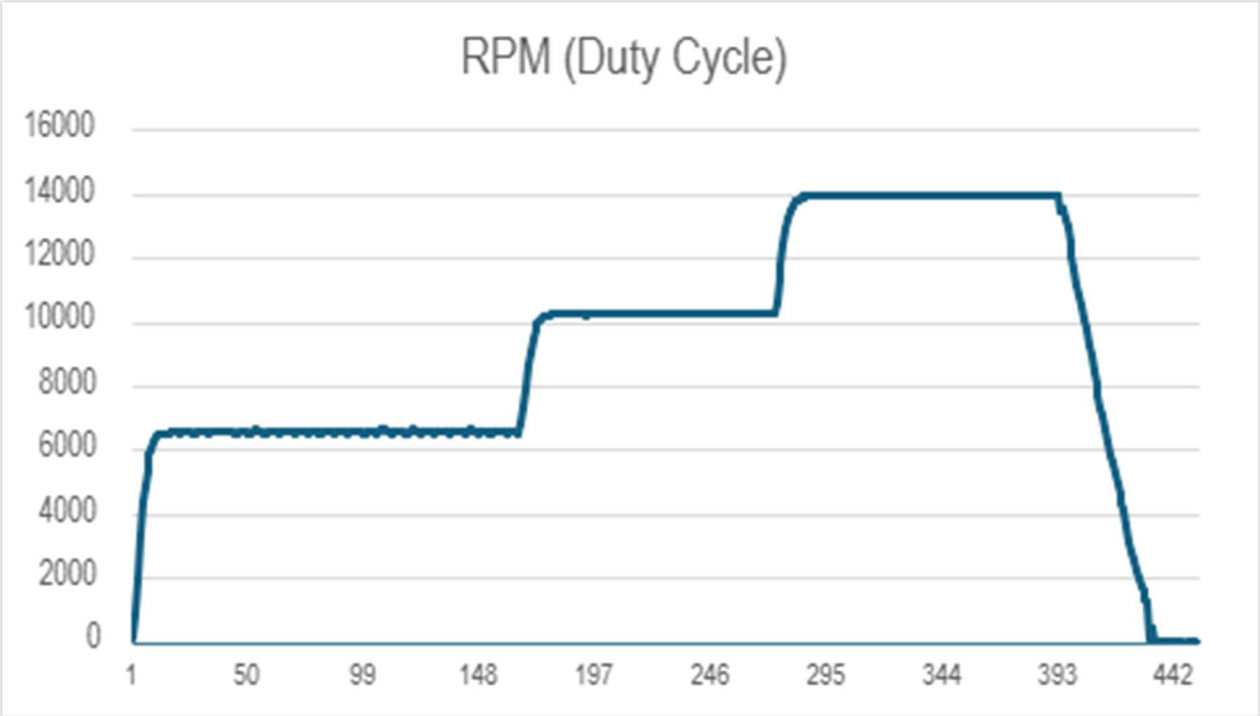
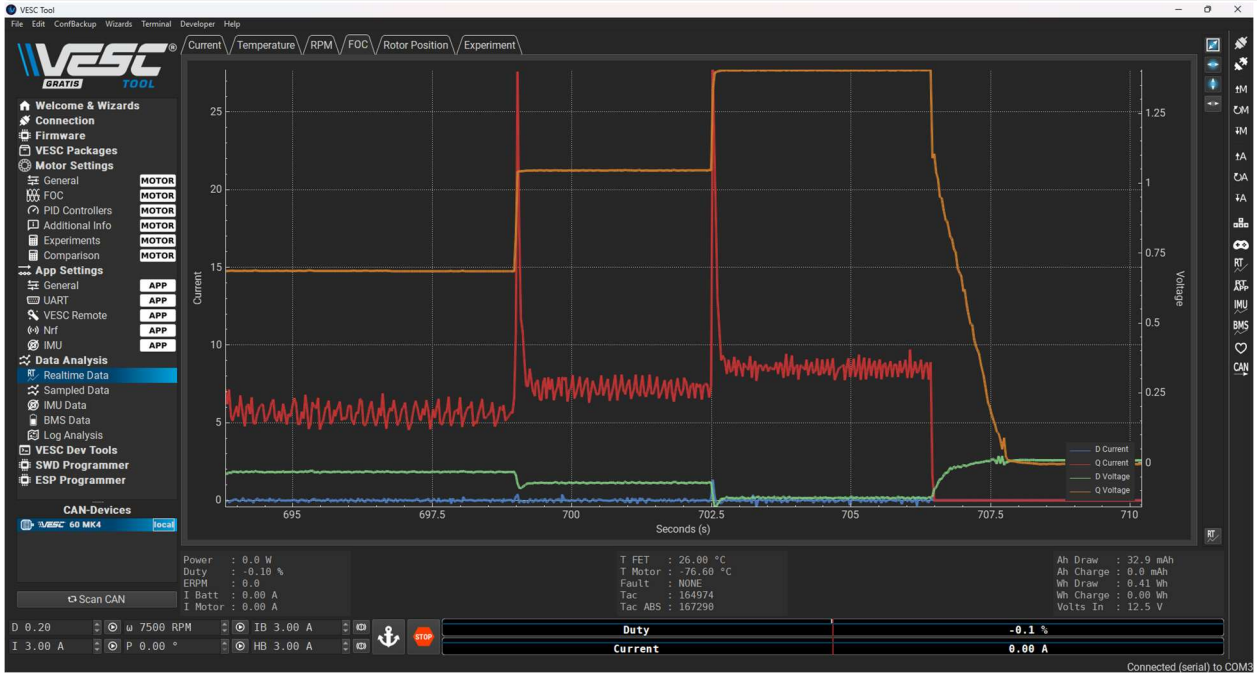

FOC Current
30
25
20
15
10
5
0
-5
-10
74363298
74366495
74369739
74373081
74376312
74379499
74382732
74385917
d_axis_current
q_axis_current



Question 4: Next test the motor operation with duty cycle input. Please maintain the duty cycle lower than 20%. (Recommended: start with a duty cycle of 0.10, then 0.15 and finally 0.2, Take a screenshot of Your RPM and FOC graphs. Log your data for each speed point and plot the graphs for RPM, FOC current (d- and q-axis currents), FOC voltage (d- and q-axis voltages)




RPM (Duty Cycle)
16000
14000
12000
10000
8000
6000
4000
2000
0
1
50
99
148
197
246
295
344
393
442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Current
Voltage
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D Current
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Power : 0.0 W
Duty : -0.10 %
ERPM : 0.0
I Batt : 0.00 A
I Motor : 0.00 A
T FET : 26.00 °C
T Motor : -76.60 °C
Fault : NONE
Tac : 164974
Tac ABS : 167290
Ah Draw : 32.9 mAh
Ah Charge : 0.0 mAh
Wh Draw : 0.41 Wh
Wh Charge : 0.00 Wh
Volts In : 12.5 V
Duty
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Current
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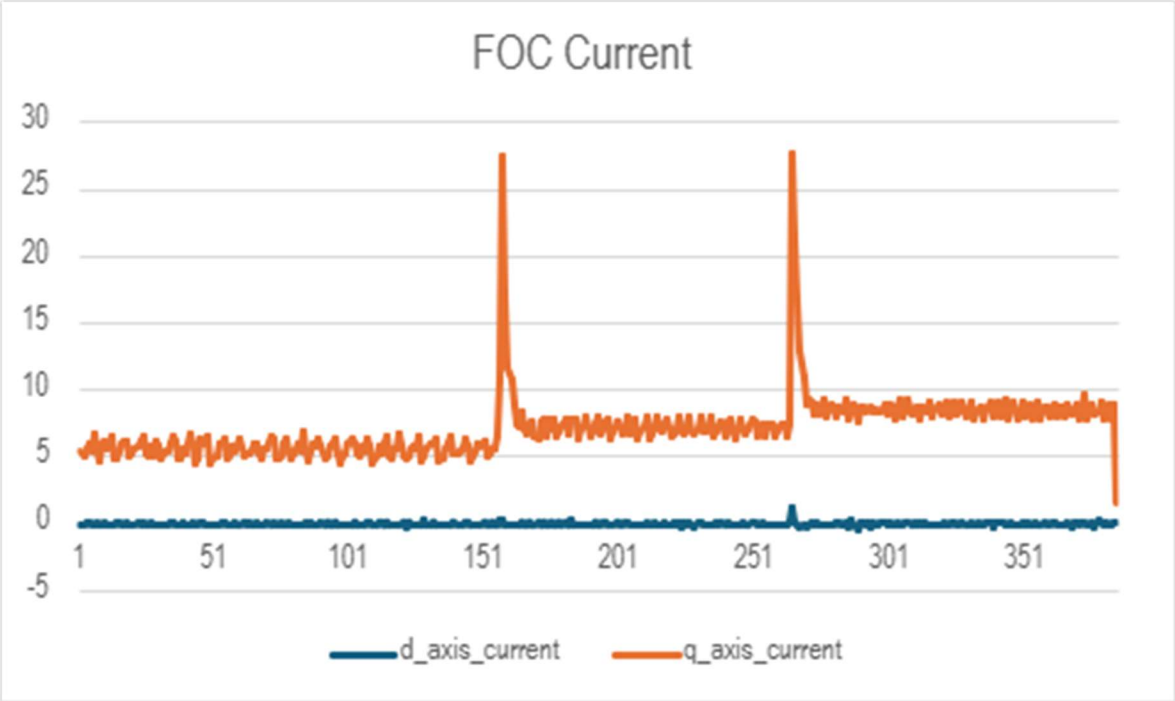
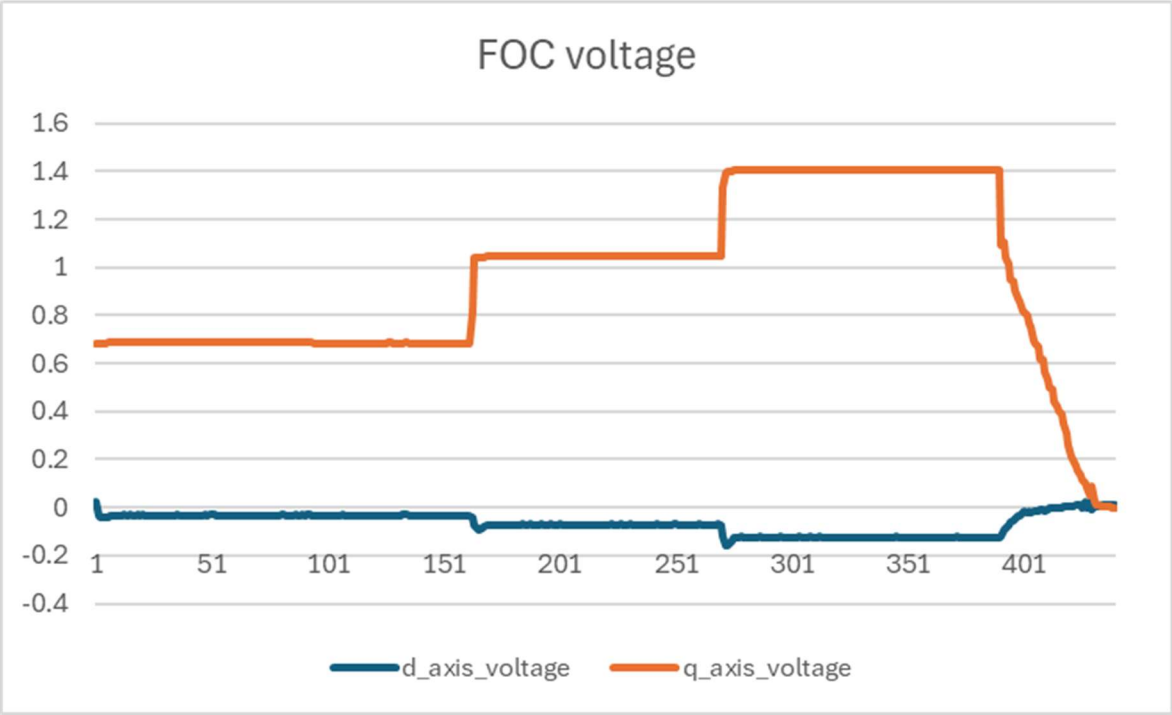

FOC Current
30
25
20
15
10
5
0
-5
1
51
101
151
201
251
301
351
d_axis_current
q_axis_current


FOC voltage
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
1
51
101
151
201
251
301
351
401
d_axis_voltage
q_axis_voltage

Question 5: What difference do you observe between the operations with the speed command and duty cycle command? Explain the difference using the real-time voltage, current, and speed waveforms

While both operations yielded comparable graphical trends, specific discrepancies were noted in the signal quality. Consistently, adjusting the **duty cycle** produced smoother waveforms than those generated by altering the speed command. At lower RPM thresholds, the speed-based control introduced significant noise; although this interference diminished as RPMs increased, the duty cycle maintained a superior level of clarity. This pattern remained uniform across all data sets, with duty cycle waveforms demonstrating a substantial reduction in noise.

Question 6: What is the definition of the d and q-axis currents? Why is the d-axis current always zero? Please check your lecture notes.

The d-axis current represents the portion of the stator current that aligns with the rotor's magnetic field and influences the motor's flux. On the other hand, the q-axis current is 90 degrees (perpendicular) to the rotor flux and produces electromagnetic torque. For a motor that runs under FOC, the rotor flux occurs due to the magnets on the motor, therefore the d-axis currrent does not increase the torque. Therefore, the d-axis current is set to zero so that all the applied current is directed into the q-axis, to maximize torque and efficiency.